Questions

Q1.

Figure 5 shows the apparatus used to investigate the melting of some crushed ice.

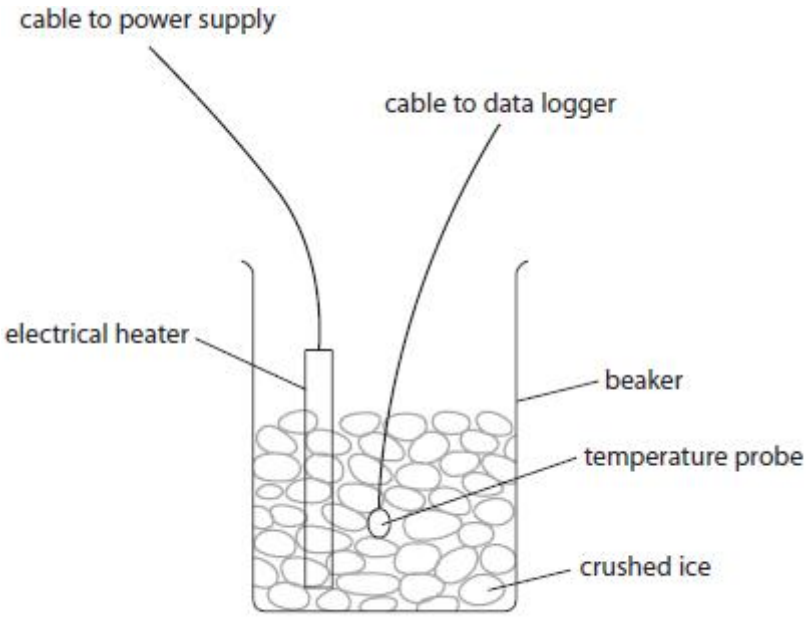

Figure 5

(i) Suggest **one** safety precaution needed when using the electrical heater.

(1)

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(ii) Suggest **one** way of heating the crushed ice without using electricity.

(1)

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(Total for question = 2 marks)

Q2.

The photograph shows a scuba diver.
She can breathe under water because she carries a cylinder of air on her back.

(i) The air molecules in the cylinder move randomly.
Describe how these air molecules exert a pressure on the cylinder.

(2)

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(ii) The cylinder contains air at a pressure of 21 000 kPa.

The volume of air in the cylinder is 15.0 litres.
When the valve on the cylinder is opened, the air expands until its pressure is 100 kPa.
The temperature of the air does not change.
Show that the new volume of air is about 3 200 litres.

(2)

(iii) The cylinder is filled with air in a hot country and then taken to a cold country.

The temperature in the hot country is 305 K.
The temperature in the cold country is 278 K.
The pressure in the cylinder in the hot country is 21 000 kPa.
Calculate the pressure in the cylinder in the cold country.

(3)

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Q3.

Figure 3 shows an oxygen cylinder.

Figure 3

The volume of the gas in the cylinder is 2100 cm^3.

When the gas is released into the atmosphere the volume of the gas is 8600 cm^3.

The pressure of the atmosphere is 98 kPa.

Calculate the pressure of the gas when it is in the cylinder.

Use the equation

$$P_1 = \frac{P_2 \times V_2}{V_1}$$

(2)

pressure of the gas in the cylinder = ... kPa

(Total for question = 2 marks)

Q4.

The photograph shows an oxygen cylinder that can be used in an ambulance.

(i) Explain how particles of oxygen gas exert a pressure on the inside of the cylinder.

(2)

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(ii) This cylinder can release 340 litres of oxygen at a pressure of 101 000 Pa.

The inside volume of the cylinder is 2.5 litres.
Use the equation

$$P_2 = \frac{P_1 V_1}{V_2}$$

to calculate the pressure of the oxygen in the cylinder before the gas is released.

(3)

pressure of oxygen = ... Pa

Q5.

The student decides to measure the temperature of the water every minute while it is being heated.

Figure 18 shows a graph of the student's results.

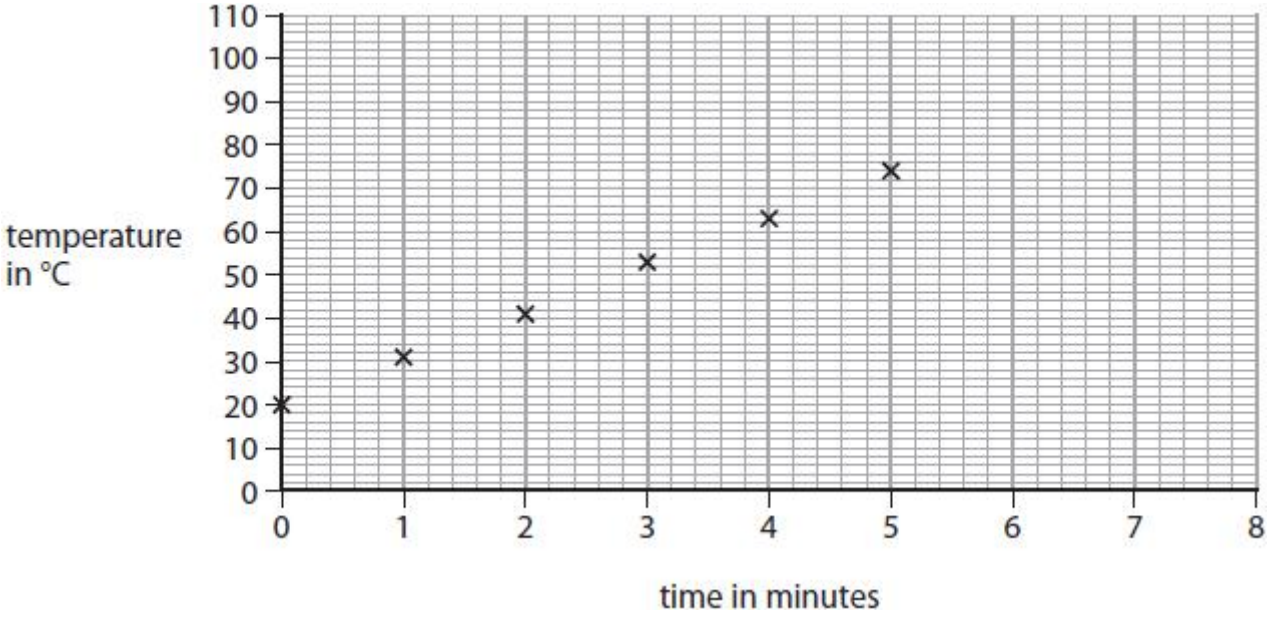

Figure 18

Predict the temperature of the water if the heating continues up to 8 minutes.

(1)

temperature of the water = ... °C

(Total for question = 1 mark)

Q6.

* A student has two metal strips and a ruler, as shown in Figure 22.

A teacher tells the student that

- one metal strip is made of aluminium
- the other metal strip is made of stainless steel.

The student looks up data in a reference book, finding some density values:

density of aluminium = 2710 kg / m^3

density of stainless steel = 7850 kg / m^3

The student has access to more of the same metal strips, if needed, and may ask for any extra measuring

devices.

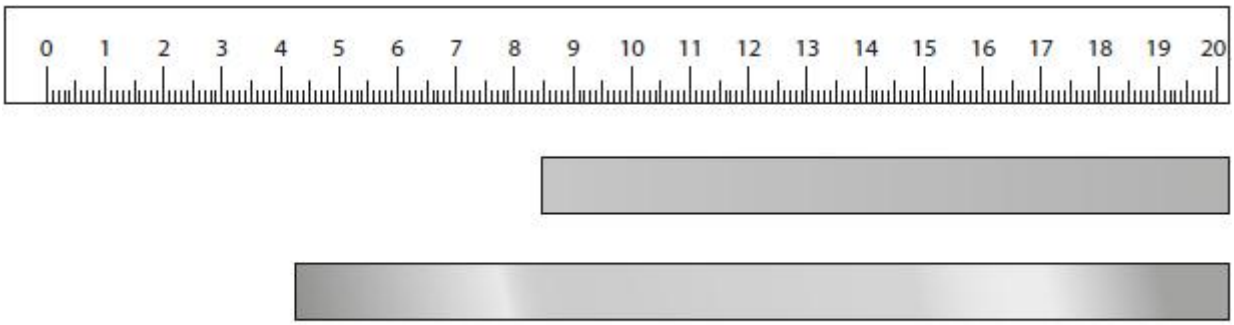

Figure 22

Plan how the student could confirm the teacher's statements, by determining the density of each of the strips as accurately as possible.

(6)

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(Total for question = 6 marks)

Q7.

A student makes the following hypothesis:

'*When I increase the pressure on a fixed mass of gas, the volume of the gas decreases.*'

She has the equipment shown in Figure 14.

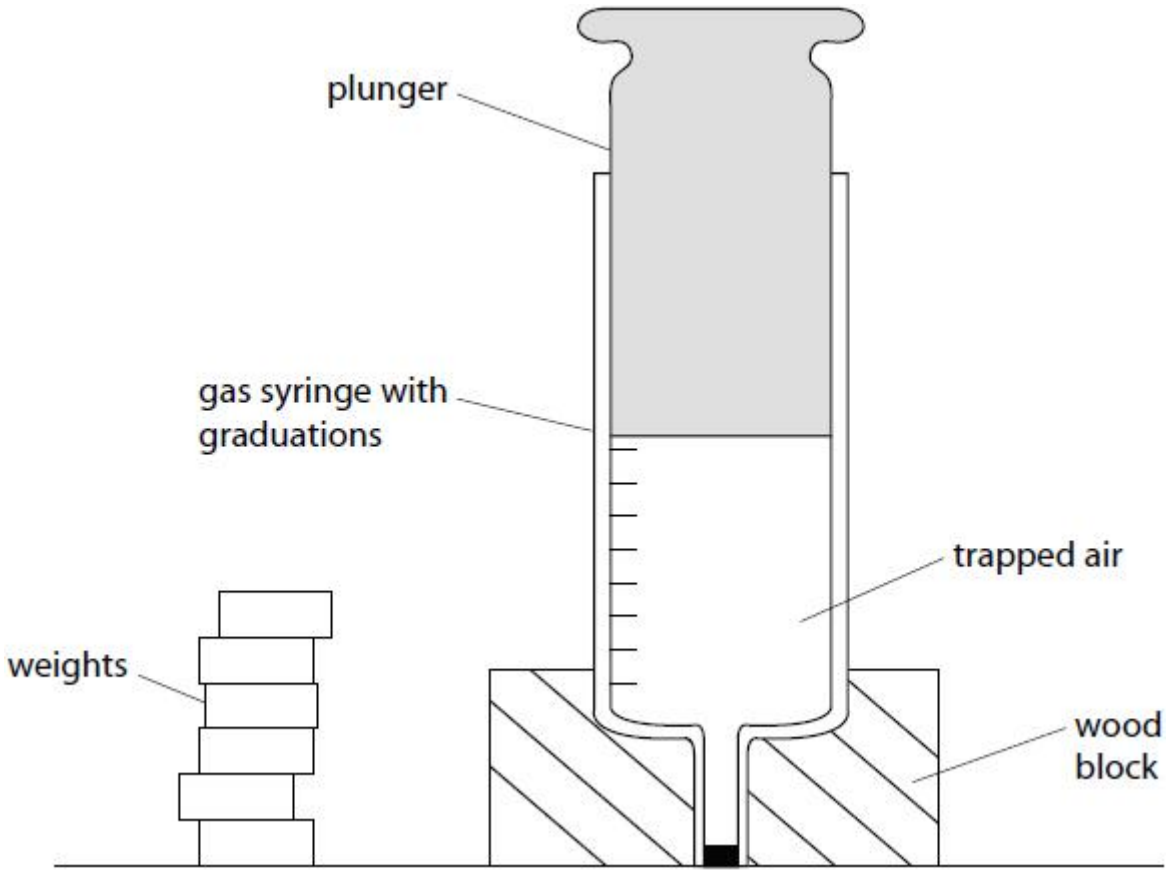

Figure 14

She measures the area of the plunger.

Devise a plan to test her hypothesis.

(4)

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(Total for question = 4 marks)

Q8.

Which row in the table is correct?

(1)

	particles in a solid	particles in a gas
☐ A	move freely	move freely
☐ B	move freely	vibrate about fixed positions
☐ C	vibrate about fixed positions	move freely
☐ D	vibrate about fixed positions	vibrate about fixed positions

(Total for question = 1 mark)

Q9.

Figure 12 shows the apparatus that is used to measure the pressure and volume of some air.

The air is trapped inside a glass tube.

The glass tube has no scale markings on it.

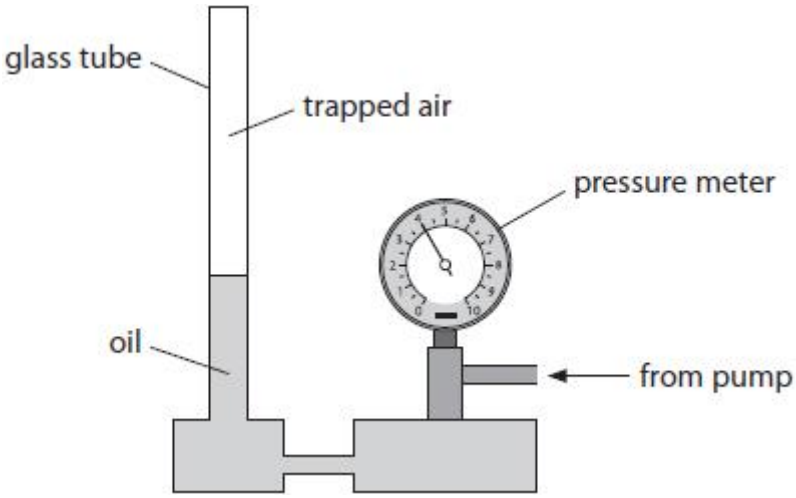

Figure 12

The volume of the trapped air is reduced by pumping oil up the glass tube.

The pressure meter is used to measure the pressure of the trapped air.

(i) Describe how you could measure the volume of the trapped air.

(3)

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(ii) * If the oil level goes down slowly, the volume of the trapped air increases and the pressure changes. The temperature remains constant.

Explain, in terms of particles, how the trapped air exerts a pressure on the inside of the glass tube and how the pressure changes as the volume increases.

(6)

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(Total for question = 9 marks)

Q10.

Kinetic theory describes the movement of particles in the three states of matter.

Gas is one of the states of matter.

(i) Name the other two states of matter.

(2)

1

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2

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(ii) Complete the sentence by putting a cross (☒) in the box next to your answer.

The average kinetic energy of the particles in a gas is directly proportional to

(1)

☐ **A** the pressure of the gas

☐ **B** the temperature of the gas measured in degrees Celsius

☐ **C** the temperature of the gas measured in Kelvin

☐ **D** the volume of the gas

Mark Scheme

Q1.

Question number	Answer	Additional guidance	Mark
i	any one from do not touch heater (when it is switched on) (1) do not use mains (voltage) (1) use gloves/cloth to handle beaker/heater (after water is heated) (1)	 accept only switch on when in beaker/water 'use gloves' must be qualified with a purpose in using them e.g. use gloves to prevent burns	**(1)** **AO1**

Question number	Answer	Additional guidance	Mark
ii	any one from bunsen (burner) (1) putting out in the sunlight (1) water bath (1)	 gas (heating) solar accept use hands (to warm)	**(1)** **AO1**

Q2.

	Answer	Acceptable answers	Mark
(i)	A description including: • collisions (1) • with (walls of) cylinder (1)	hit / bounce off exert force	**(2)**
(ii)	substitution (1) either 100 × V = 15.0 × 21 000 or $V = \frac{15.0 \times 21\,000}{100}$ evaluation (1) 3 150(litres)	V_1P_1 = 15 × 21000 = 315000 (1 mark) V_2P_2 = **100** × 3200 = 320000 (1 mark) award full marks for 3150 (litres) without working	**(2)**
(iii)	substitution (1)	give full marks for	**(3)**

	$\frac{21\,000\ (\times V)}{305} = \frac{P\ (\times V)}{278}$ volume same (1) evaluation (1) 19 100 (kPa)	correct answer, no working transposition accept 19141 (kPa) or 19000 and numbers rounding down to 191 00	

Q3.

Question Number:	Answer	Additional guidance	Mark
	substitution (1) $(P_1) = \frac{98 \times 8600}{2100}$ evaluation (1) 400 (kPa)	 accept any answer that rounds to 400 eg. 401.33 (kPa) award full marks for the correct answer without working.	**(2)** AO 2 1

Q4.

Question Number	Answer	Acceptable answers	Mark
(i)	an explanation **linking** two of the following three points:- particles move (1) bombarding/colliding (1) with wall/side (1) (only give if one of the previous marks is there) (of container)	 molecules/they move hit ignore 'pushing' e.g. molecules push on walls = 0 bounce off inside of container =2	**(2)**

Question Number	Answer	Acceptable answers	Mark
(ii)	substitution $P_2 = \frac{101\ 000 \times 340}{2.5}$ (1) Evaluation 13.7 to any power of 10 (1) 13 700 000(Pa), 13 700kPa (1)	 1.37(36) X 10^7/ 13736000 14 to any power of 10 14 000 000 (Pa), 14 000 (kPa) Full marks are awarded for the correct answer with no working	**(3)**

Q5.

Question Number:	**Answer**	**Additional guidance**	**Mark**
	100 (°C) (1)	accept any answer between and including 95 and 102 (possibility that it is not pure water and possibility of heat loss prevents reaching boiling point)	**(1)** AO 2 1

Q6.

Question number	Indicative content	Mark
*	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. **AO1 (strand 2) (6 marks)** Indicative content • measure the length and width of a strip with the ruler / a metre rule • measure the thickness of the strip with a more accurate device e.g. digital callipers OR place 5 (say) of the same strip on top of each other and measure their thickness with the ruler then ÷5 to calculate a single thickness [plus air gap] • measure the mass of a strip with an electronic balance • measure the mass of (say) 5 strips then ÷5 to calculate the mass of one of them • calculate the volume (= l x w x t) in m^3 and the mass in kg • use displacement can/measuring cylinder to find the volume • mass / volume to get density • check if it's near one of the teacher's two values of density given • if it's close / not so far off it's safe to assume that strip is of the identified material • repeat for the other strip • other repeat measurements	**(6)** **AO1**

AO targeting: AO1.2

Level	Mark	Descriptor
	0	• No rewardable material.
Level 1	1-2	• Demonstrates elements of physics understanding, some of which is inaccurate. Understanding of scientific, enquiry, techniques and procedures lacks detail. (AO1) • Presents a description which is not logically ordered and with significant gaps. (AO1)
Level 2	3-4	• Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas, enquiry, techniques and procedures is not fully detailed and/or developed. (AO1) • Presents a description of the procedure that has a structure which is mostly clear, coherent and logical with minor steps missing. (AO1)
Level 3	5-6	• Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas, enquiry, techniques and procedures is detailed and fully developed. (AO1) • Presents a description that has a well-developed structure which is clear, coherent and logical. (AO1)

Summary for guidance

Level	Mark	Additional Guidance	General additional guidance – the decision within levels e.g. - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level.
	0	No rewardable material.	
Level 1	1–2	Additional guidance Partially complete description of a suitable procedure with at least two measurements OR one measurement and another procedural point	Possible candidate responses measure the length measure the width of a strip measure the mass/weight of a strip e.g. repeat measurements
Level 2	3–4	Additional guidance Mostly complete description of a suitable procedure with at least three measurements and some description of processing the results.	Possible candidate responses As above with measure the thickness of the strip calculate the volume (= l x w x t) OR immerse in liquid to get volume
Level 3	5–6	Additional guidance Detailed description of a suitable procedure with all necessary measurements and a clear description of processing the results.	Possible candidate responses As above with extra detail e.g. measure the mass of (say) 5 strips then ÷5 to calculate the mass of one of them detail of obtaining volume by immersion use density =mass /volume check if density value obtained is near one of the teacher's two values

Q7.

Question number	Answer	Mark
	An answer that combines the following points to provide a plan: • put weights on the plunger to increase the pressure of the trapped air (1) • use scale on syringe to measure the volume of trapped air (1) • calculate the pressure from P = weight added/area of plunger (1) • compare the increase in pressure to the volume of trapped air (1)	(4)

Q8.

Question number	Answer	Mark
	C vibrate about fixed positions move freely	**(1)**

Q9.

Question number	Answer	Additional guidance	Mark
(i)	An answer that combines the following points of application of knowledge and understanding to provide a logical description of a method: measure the length of the air column (1) use a rule or fixed scale (1) use volume = area x length (1)	 measure (/find known) area of tube	**(3)**

Question number	Indicative content	Mark
* (ii)	Answers will be credited according to the candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all of the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. **A02 (6 marks)** • particles in (constant / random) motion • collide with the glass / oil surface • exert a force on the glass /change in momentum • as volume increases, pressure decreases • collisions with glass less frequent • same number of collisions over larger area • same force over larger area • (means) lower pressure	(6)

Level	Mark	Descriptor
	0	No rewardable material.
Level 1	1–2	The explanation attempts to link and apply knowledge and understanding of scientific ideas, flawed or simplistic connections are made in the context of the question (A02) Lines of reasoning are unsupported or unclear (A02)
Level 2	3–4	The explanation is mostly supported through linkage and application of knowledge and understanding of scientific ideas. Some logical connections made between elements in the context of the question. (A02) Lines of reasoning are mostly supported through the application of relevant evidence (A02)
Level 3	5–6	The explanation is supported throughout by linkage and application of knowledge and understanding of scientific ideas. Logical connections made between elements in the context of the question. (A02) Lines of reasoning are supported by sustained application of relevant evidence (A02)

Q10.

Question Number	Answer	Acceptable answers	Mark
(i)	solid liquid	in either order plasma as an alternative to either.	**(2)**

Question Number	Answer	Acceptable answers	Mark
(ii)	**C** temperature of the gas measured in Kelvin		**(1)**